

Reg. No:



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme : B. E / B. tech

Course Code : CSA4301

Course Name: Internet programming

Problem Statement (Assignment)

Max Marks:

100

Field	Content
Assignment Title	Smart Campus Service Management System
Course Outcome(s) – CO	CO1: Develop visually appealing web pages using CSS for layout and style, and create dynamic, interactive functionality with JavaScript. CO2: Build dynamic web applications using XML, XSLT, and JSP within the MVC paradigm for efficient data handling and presentation.
Bloom's Taxonomy Level	L3 – Apply L4 – Analyze L5 – Evaluate L6 – Create
SDG Mapping (SDG 1–17)	SDG 9 - Industry, Innovation and Infrastructure, SDG 4 - Quality Education

Problem Statement

A university currently relies on manual communication such as emails, phone calls, and paper-based records for reporting and resolving campus service issues. This approach makes it difficult to efficiently record complaints, assign requests to the appropriate service personnel, monitor progress, and provide timely status updates to students and faculty.

The university therefore proposes to develop a Smart Campus Service Management System as an integrated web-based application. The system shall enable students and faculty to submit service requests related to laboratory equipment faults, network problems, classroom maintenance,

library services, and other technical issues through a web browser. Users should be able to provide relevant details, submit requests, and subsequently track the status of their requests.

The proposed system should demonstrate an integrated client-server web architecture. The client side shall provide an interactive and user-friendly interface using HTML/XHTML, CSS, JavaScript, and DOM. The server side shall process requests using Java Servlets and JSP, following the MVC architecture. XML shall be used for structured data representation, and XSLT shall be used to transform XML data for presentation where appropriate.

Problem Identification (10 Marks)-The university relies on manual communication (emails, phone calls, paper records) for reporting and resolving service issues, which causes inefficiency in recording complaints, assigning requests, monitoring progress, and providing timely updates.

Proposed Solution -Develop a **Smart Campus Service Management System (10 Marks)**, a web-based application that allows students and faculty to submit service requests (lab equipment faults, network problems, classroom maintenance, library services) and track their status online.

System Design & Architecture - Data Representation (20 Marks) -Implement an integrated **client-server web architecture**. The client side should use HTML/XHTML, CSS, JavaScript, and DOM for interactivity, while the server side should use Java Servlets and JSP following the **MVC model**. Use **XML** for structured data representation for transforming XML data into user-friendly formats for presentation and reporting.

Implementation & Testing (10 Marks) -Demonstrate request submission, assignment to service personnel, progress tracking, and timely updates. Validate inputs, handle errors, and ensure smooth user experience.

STG Goals –**SDG 9** - Industry, Innovation and Infrastructure, **SDG 4** - Quality Education

Requirement Analysis (*SDG 9 – Industry, Innovation & Infrastructure*) **Identify inefficiencies in manual reporting and propose a digital infrastructure for service management.**

Client Interface Development (*SDG 4 – Quality Education*) **Build user-friendly web forms and dashboards so students/faculty can easily submit and track requests, improving access to campus services.**

Assessment Criterion	CO Mapping	Weightage / Marks
Problem understanding & formulation	CO1	10
Application of course/domain knowledge	CO2	20
Solution methodology / design / implementation	CO3	20
Use of appropriate modern tools / techniques	CO4	10
Results, testing & validation	CO4 / CO5	15
Analysis, trade-offs & justification	CO5	15
Broader considerations / professional responsibility, where applicable	CO5	5
Technical documentation & reflection	CO1–CO5	5
TOTAL		100

Performance Level	CO Attainment Percentage	Interpretation
Level 3	≥ 70%	CO Attained
Level 2	60–69%	CO Moderately Attained
Level 1	50–59%	CO Minimally Attained
Level 0	< 50%	CO Not Attained

Code of Conduct

I _____ **CH.Bhavya Sri(192411249)**_____ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic-integrity rules will result in disciplinary action.

Signature of the Student: **CH.Bhavya Sri**_____

Faculty In-charge Name: Date:	Course Coordinator Name: Date:	Head of Department Name: Date:
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TITLE:SMART CAMPUS SERVICE MANAGEMENT SYSTEM

1. Problem Understanding and Formulation

1.1 Problem Statement:A university currently depends on manual communication methods such as emails, phone calls, and paper-based records to report and resolve campus service issues. This makes the process slow and difficult to manage. Complaints may not be recorded properly, requests may not reach the appropriate service personnel, and students and faculty may not receive timely updates.

The proposed **Smart Campus Service Management System** provides an integrated web-based solution through which students and faculty can submit service requests and track their progress online.

The system covers issues such as:

Laboratory equipment faults

Network problems

Classroom maintenance

Library services

Computer and technical problems

Other campus-related services

1.2 Existing System

The existing system mainly depends on:

Emails

Phone calls

Direct communication

Paper records

Manual assignment of complaints

1.3 Problems in Existing System

The manual approach results in:

Delay in complaint registration

Difficulty maintaining records

Improper assignment of requests

Lack of request tracking

Delayed status updates

Difficulty generating reports

Increased possibility of missing or duplicate requests

1.4 Objective

The main objective is to develop a centralized web-based system that allows users to report campus service issues, enables administrators to assign requests, and allows service personnel to update progress.

2. Application of Course/Domain Knowledge

The project applies concepts from web programming, server-side programming, data representation, and software architecture.

2.1 HTML/XHTML

HTML/XHTML is used to create the structure of the web pages.

Examples:

Login page

Registration page

Service request form

User dashboard

Request tracking page

2.2 CSS

CSS is used to improve the appearance and layout of the application.

It provides:

Page layout

Forms

Buttons

Tables

Navigation menus

Responsive design

2.3 JavaScript

JavaScript provides client-side interactivity.

It can be used for:

Input validation

Dynamic messages

Form checking

Status display

Interactive dashboard elements

2.4 DOM

The **Document Object Model (DOM)** allows JavaScript to access and modify webpage elements dynamically.

For example, when a user leaves a required field empty, JavaScript can use DOM to display an error message without reloading the complete page.

2.5 Java Servlets

Java Servlets act as server-side controllers. They receive requests from the browser, process them, and send appropriate responses.

Example:

Submit Request → Servlet → Process Request → Response

2.6 JSP

JavaServer Pages (JSP) are used to create dynamic web pages.

JSP can display:

User information

Service requests

Request status

Assigned personnel

Reports

2.7 MVC Architecture

The application follows the **Model-View-Controller (MVC)** architecture.

Model

Manages data and business logic.

View

Provides the user interface using JSP/HTML.

Controller

Handles user requests using Java Servlets.

MVC Flow

User → View → Controller → Model → Controller → View → User

2.8 XML

XML is used for structured representation of service request information.

Example:

```
<ServiceRequest>
  <RequestID>SR001</RequestID>
  <Category>Network</Category>
  <Location>Computer Lab</Location>
  <Description>Internet connection unavailable</Description>
  <Priority>High</Priority>
  <Status>Submitted</Status>
</ServiceRequest>
```

2.9 XSLT

XSLT is used to transform XML data into a user-friendly presentation format such as HTML reports.

XML → XSLT → HTML Report

3. Solution Methodology / Design / Implementation – CO3 – 20 Marks

3.1 Proposed Solution

The proposed Smart Campus Service Management System provides a centralized platform for campus service management.

The basic workflow is:

User Login



Submit Service Request



Servlet Processes Request



Request Stored



Administrator Reviews Request

↓

Request Assigned to Service Personnel

↓

Service Personnel Updates Status

↓

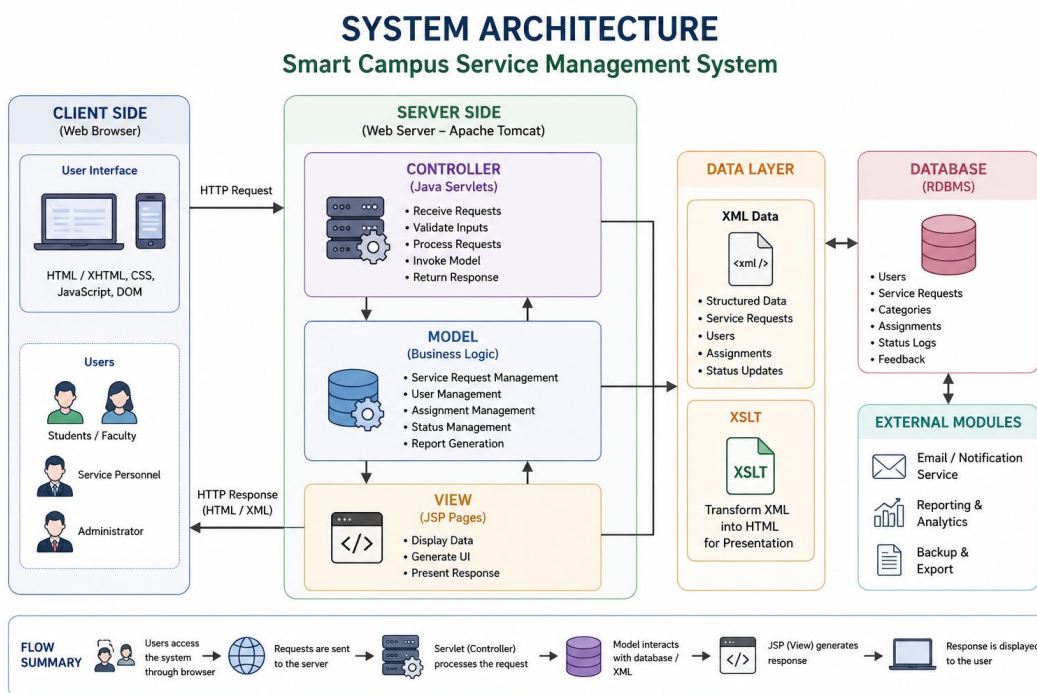
User Tracks Status

↓

Request Resolved and Closed

3.2 System Architecture

The system uses an integrated client-server architecture.



3.3 System Modules

Module 1 – User Management

Provides:

Registration

Login

Logout

User profile

Authentication

Module 2 – Service Request Management

Users can:

Select service category

Enter location

Describe the problem

Select priority

Submit the request

Module 3 – Request Assignment

The administrator reviews requests and assigns them to appropriate service personnel.

Module 4 – Request Tracking

Users can view:

Request ID

Category

Description

Assigned personnel

Current status

Resolution details

Module 5 – Status Management

Service personnel can change the request status:

Submitted → Assigned → In Progress → Resolved → Closed

Module 6 – Reporting

The system can generate reports from structured XML data.

Examples:

Pending requests

Completed requests

Category-wise requests

Personnel-wise requests

Request history

3.4 Sample Request

A student submits:

Request ID: SR001

Category: Network Problem

Location: Computer Lab 2

Description: Internet connection is not working.

Priority: High

Status: Submitted

The administrator assigns it to a network technician.

The technician updates:

Submitted → Assigned → In Progress → Resolved → Closed

The student can track the status through the dashboard.

4. Use of Appropriate Modern Tools / Techniques – CO4 – 10 Marks

The project uses appropriate technologies for developing an integrated web application.

Tool/Technology	Purpose
HTML/XHTML	Webpage structure
CSS	Styling and layout
JavaScript	Client-side interaction
DOM	Dynamic webpage manipulation
Java	Server-side programming
Java Servlets	Request processing/controller
JSP	Dynamic web pages
MVC	Application architecture
XML	Structured data representation
XSLT	XML transformation
Apache Tomcat	Web application server
Web Browser	Client-side access

These technologies are suitable because they demonstrate both client-side and server-side web development concepts required for the project.

5. Results, Testing and Validation – CO4 / CO5 – 15 Marks

5.1 Expected Result

The developed system should allow a student or faculty member to:

Login to the system.

Submit a service request.

Receive a request ID.

Track the request.

View status updates.

Receive the final resolution status.

Administrators should be able to:

View submitted requests.

Assign requests.

Monitor progress.

Generate reports.

Service personnel should be able to:

View assigned requests.

Update progress.

Mark requests as resolved.

5.2 Testing

Test Case	Input	Expected Result
TC01	Valid login	Dashboard displayed
TC02	Invalid login	Error message displayed
TC03	Valid service request	Request submitted
TC04	Empty required field	Validation message
TC05	Request assignment	Personnel assigned
TC06	Status update	New status displayed
TC07	Request tracking	Request details displayed
TC08	Invalid request data	Error handled
TC09	XML data	Structured data processed
TC10	XSLT transformation	Readable report generated

5.3 Validation

Input validation is performed to ensure:

Required fields are completed.

Correct user credentials are entered.

Valid service categories are selected.

Meaningful descriptions are provided.

Unauthorized users cannot access restricted functions.

Error messages are displayed whenever invalid information is entered.

6. Analysis, Trade-offs and Justification

6.1 Comparison with Existing System

Existing Manual System	Proposed System
Paper/email based	Web based
Manual complaint recording	Digital recording
Manual assignment	Centralized assignment
Difficult tracking	Online tracking
Delayed updates	Timely status updates
Difficult reporting	XML/XSLT-based reports
Records may be difficult to search	Centralized records

6.2 Advantages

Reduces paperwork.

Saves time.

Centralizes service requests.

Improves request assignment.

Provides request tracking.

Improves transparency.

Simplifies report generation.

Provides better communication.

6.3 Trade-offs

The web-based system provides better efficiency than manual communication, but it requires an operational server, network connectivity, and proper user authentication.

Using multiple technologies such as Servlets, JSP, XML, and XSLT provides a clear demonstration of course concepts but can make development more complex than using a single modern web framework.

Justification

The selected technologies are appropriate because they directly demonstrate the required concepts of client-server architecture, MVC, dynamic web pages, structured data representation, and XML transformation.

7. Broader Considerations / Professional Responsibility

7.1 SDG 9 – Industry, Innovation and Infrastructure

The project supports **SDG 9** by replacing inefficient manual reporting with a digital infrastructure for campus service management.

The system:

Identifies inefficiencies in manual reporting.

Provides centralized digital infrastructure.

Improves service management.

Promotes digital transformation.

Supports efficient use of university resources.

7.2 SDG 4 – Quality Education

The project supports **SDG 4** by making campus services easier for students and faculty to access.

The system provides:

User-friendly service request forms.

Online request tracking.

Faster reporting of classroom and laboratory issues.

Improved access to technical and library services.

Better learning-environment support.

7.3 Security and Responsibility

The system should protect user information through:

Authentication

Authorization

Input validation

Secure access to administrator functions

Proper handling of user data

8. Technical Documentation and Reflection – CO1–CO5 – 5 Marks

8.1 Technical Documentation

The project documentation describes:

Problem statement

Objectives

Requirements

System architecture

Technologies used

Modules

Workflow

Implementation

Testing

Results

SDG contribution

Future enhancements

The documentation provides a clear explanation of how the system is designed and implemented.

8.2 Personal Reflection

Through this project, I gained practical understanding of client-server web application development. I learned how HTML, CSS, JavaScript, DOM, Java Servlets, and JSP work together to create an interactive web application. I also understood the importance of MVC architecture in separating the user interface, request processing, and business logic.

Working with XML and XSLT helped me understand structured data representation and data transformation. Testing the system improved my understanding of input validation, error handling, and software reliability.

The project also helped me understand how technology can solve real-world problems in an educational environment and how software projects can contribute to broader goals such as **SDG 9** and **SDG 4**.

9. Future Enhancements

The system can be improved in the future by adding:

Email/SMS notifications

Mobile application

AI-based request prioritization

Automatic personnel assignment

Image/file attachment

QR-code-based complaint reporting

Analytics dashboard

User feedback and rating

Cloud deployment

Advanced reporting

10. Conclusion

The **Smart Campus Service Management System** provides a digital solution for efficiently managing university service requests. It replaces manual communication methods with an integrated web-based platform that allows students and faculty to submit and track service requests.

The system demonstrates important course concepts including **HTML/XHTML, CSS, JavaScript, DOM, Java Servlets, JSP, MVC, XML, and XSLT**. The proposed architecture separates the client interface, server-side processing, business logic, and data representation.

The system improves the efficiency, transparency, and accessibility of campus service management. It also supports **SDG 9 – Industry, Innovation and Infrastructure** by introducing digital infrastructure and **SDG 4 – Quality Education** by improving access to essential campus services.